Online

General Biology (Biology One).

[01-20-26]

Metric Measurement

↳ Four base units:
- Grams (g) For Weight
- Meters (m) For Length
- Liters (L) For Volume
- Celsius (°C) For Temperature

Metric Rulers show numbers as centimeters, with ten marks between each number. Each smaller mark is a millimeter.

↳ A centimeter (cm) is 100 times smaller than a meter. A Millimeter (mm) is 1,000 times smaller than a meter.

Example: 6.7cm = 67mm

The base metric unit for temperature is 1°C (one degree celsius).

↳ At 0°C, water will freeze.
At 100°C, water will boil.

The Base metric unit for Volume is the Liter.

↳ A MilliLiter (mL) is 1,000 times smaller than a liter

Water adheres to the glass walls of a graduated cylinder, creating a curved appearance termed a meniscus. Volume measurements are taken at the bottom of the meniscus.

The base metric unit for weight or mass is the gram (g).
↳ A milligram (mg) is 1,000 times smaller than a gram.
↳ A Kilogram is 1,000 times larger than a gram

Mass is the amount of matter an object contains. Weight is a measure of the force of gravity on an object or mass

Metric Conversions Quiz Work

1. 10cL to mL: 1cL = 10mL
↳ 10cL × 10mL/cL = 100mL

2. 6Km to m: 1 Km = 1,000m

$6km \times \frac{1{,}000m}{1\ Km}$ → 6 x 1,000m → 6 x 1,000 = 6,000 → 6,000 Km.

3. 120mm to cm → 1cm = 10mm → $120mm \times \frac{1cm}{10mm}$
→ $\frac{120}{10}$cm → 120 ÷ 10 = 12cm

4. 5g to Kg: 1kg = 1,000g → $5g \times \frac{1Kg}{1{,}000g}$
→ $\frac{5}{1000}$Kg → 5 ÷ 1000 = 0.005 Kg

5. 1 Km to miles: 1 Km = 0.62 mi →
450 Km X 0.62 $\frac{miles}{Km}$ → 450 x 0.62 ≈ 279.6

6. 16 mL to inches: 1 inch = 25.4mm →
$16mm \times \frac{1in}{25.4mm}$ → $\frac{16}{25.4}$ in → 16 ÷ 25.4 = 0.629
→ 16mm ≈ .63 in

7. 2L to gallons: 1 liter = 0.26 gal →
2L X 0.26 $\frac{gal}{L}$ → 2 x 0.26 = 0.52 gal

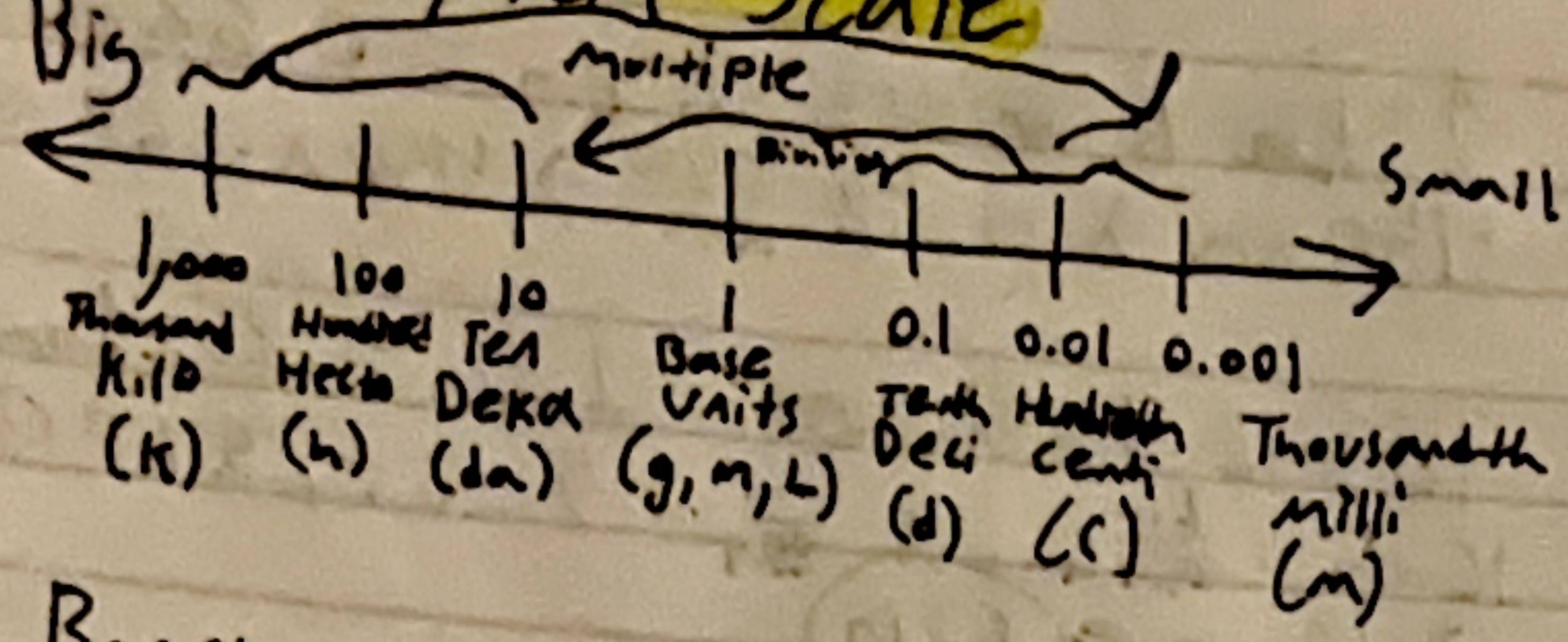
8. 180Lbs to kg: 1 pound = 0.45kg →
180lb X 0.45 $\frac{kg}{lb}$ → 180 X 0.45 = 81

9. 0.525m to in: 1m = 39.37 in → 0.525m X 39.37 $\frac{in}{m}$ → 0.525 X 39.37 = 20.66

10. 355mL to (Oz X 6): 1 mL = 0.03 fl oz →
355mL X 0.03 $\frac{fl\ oz}{mL}$ → 355 x 0.03 = 11.9 fl oz →
12 X 6 = 72 fl oz

11. 2lb/week x 4 weeks = 8 lb month
1 lb = 0.45 kg → 8 lb x 0.45 kg = 3.62 kg

Metric System Scale



Based on number ten.

K h dk $\boxed{\text{Meters Liters Grams}}$ d c m

Move decimal right or multiply

Move decimal left or divide

Dimensional Analysis (Factor-labeling)

↳ Converts from one set of measurements to another set of measurements. Requires the use of equivalent factors (equal amounts) given in a conversion table.

Units of Measure

Length

- 1 meter (m) = 1,000 millimeters (mm)
- 1 meter = 100 centimeters (cm)
- 1 meter = 10 decimeters (dm)
- 1 dekameter (dam) = 10 meters
- 1 hectometer (hm) = 100 meters
- 1 kilometer (km) = 1,000 meters

- 1 foot (ft) = 12 inches (in)
- 1 yard (yd) = 3 feet
- 1 yard = 36 inches
- 1 mile (mi) = 1,760 yards
- 1 mile = 5,280 feet
- 1 inch = 2.54 centimeters

Capacity

- 1 liter (L) = 1,000 milliliters (mL)
- 1 liter = 100 centiliters (cL)
- 1 liter = 10 deciliters (dL)
- 1 dekaliter (daL) = 10 liters
- 1 hectoliter (hL) = 100 liters
- 1 kiloliter (kL) = 1,000 liters

- 1 cup (c) = 8 fluid ounces (fl oz)
- 1 pint (pt) = 2 cups
- 1 quart (qt) = 2 pints
- 1 quart = 4 cups
- 1 gallon (gal) = 4 quarts
- 1 gallon = 3.784 liters

Weight/Mass

- 1 gram (g) = 1,000 milligrams (mg)
- 1 gram = 100 centigrams (cg)
- 1 gram = 10 decigram (dg)
- 1 dekagram (dag) = 10 grams
- 1 hectogram (hg) = 100 grams
- 1 kilogram (kg) = 1,000 grams

- 1 pound (lb) = 16 ounces (oz)
- 1 ton (T) = 2,000 pounds
- 1 pound = 453.59 grams
- 1 kilogram = 2.204 pounds

Chapter One (The Study of Life).

Notes - Color Coding:

- Green - Vocab terms (blue in book)
- Pink - Scientists (purple in book)
- Orange - theories (green in book)
- Yellow - steps/processes (red in book)

Biology: A natural science concerning the study of life and living organisms, including their structure, function, growth, origin, evolution, distribution, and taxonomy ↳ bio = life, -logy = science/study of.

↳ Many branches:

- Zoology → study of animal life
- MicroBiology → study of micro-organisms
- Anatomy → study of the structure of living organisms
- Botany → Study of plant life
- Ecology → study of organisms, their environment
- Physiology → study of animal function

Life is Diverse

↳ Range in size from microscopic bacteria to 300ft tall sequoia trees and 40-ton humpback whales.

Estimated 8.7+ million species on the planet excluding bacteria. Bacteria would double that number.

Life is defined by characteristics of life which distinguish living organisms from non-living things:

- Organized
- Requires Energy
- Maintains Homeostasis
- Adapts and Evolves.
- Responds to Stimuli
- Reproduces and Develops

Life is Organized

↳ Inside an organism: Cells → Tissues → Organs → Organ systems → Organism

↳ Outside the organism: Individual → population → Community → Ecosystem → Biome → Biosphere.

Life Requires Energy

- it takes work and energy to maintain the organization of cell and organism.

- Energy → the capacity to do work.
- Photosynthesis → in producers, use of solar energy to produce chemical energy (Food).
- Metabolism → in all organisms, chemical reactions convert food into energy.

Life Maintains Homeostasis → A state of biological balance. Homeo- = Same, steady.
-stasis = Controlling, constant.

To survive, organisms must keep certain physiological factors (temperature, moisture level, acidity, etc.) within a constant, tolerable level.
↳ Ex: The nervous system controls breathing rate to ensure your body is receiving the appropriate amount of oxygen.
The body maintains a steady temperature, through processes such as sweating/shivering.

Life responds to stimuli – Chemical or physical change in internal or external environment, causes a response from the organism. Closely connected to homeostasis.
Ex: Moths attracted to light → light is a stimulus
Plants grow towards light

Life Reproduces and Develops
• Reproduction → producing new individual of same kind. Genetic information (DNA) is passed to new individual.
• Development → process of growth and differentiation of cells and tissues. Each new cell is transformed through series of stages.

Life Adapts and Evolves
Adaption → modifications that make one organism better fit to function and survive in a particular environment.

Evolution – the way a population changes over many generations to become more suited to their environment.

Natural Selection
↳ Specific characteristic → better suited for environment → survive and reproduce.

Organisms can survive and pass this characteristic on to future generations. Survival of the fittest.

Charles Darwin (1809 – 1882)
↳ English naturalist. Laid foundation of evolution based on his theory of natural selection. Formulated his theory in 1839 after voyaging around the world on his ship HMS Beagle. He published his theory in 1859 in his book "On the Origin of Species."

Greatest Contribution → Theory of Evolution:
• Individuals of species are not identical.
• Traits are passed down from generation to generation
• More offspring are born than can survive
• Only survivors of competition for resources will reproduce

Alfred Wallace (1823 – 1913)
↳ British naturalist, explorer, geographer, biologist, and anthropologist. Proposed theory of evolution by natural selection in 1858. Prompted Darwin to present his ideas. Wallace and Darwin co-authored a →

→ publication in 1858. The Success of "On the Origin of Species." overshadowed Wallaces' Contributions.

Greatest Contributions: 8 year excursion to Dutch East Indies (Indonesia), developed natural selection theory, discovered thousands of new species of beetles.

Classification of Life

Due to the diversity of life it is helpful to group organisms into Categories.

Taxonomy - study of naming, describing and classifying organisms. Taxo- = arrange, -nomy = usage.

Systematics - study of evolutionary relationships and shared traits

Carl Linnaeus (1707-1778)

↳ Swedish botanist, zoologist, taxonomist, and physician. Originator of modern classification system. Named 4,400 animal species and 7,700 plant species. Authored over 180 works, but most notable is his "Systema Naturae."

Greatest Contributions: Binomial Nomenclature → A naming system in which each species is assigned a two-part name, Developed heirarchal system for classifying organisms.

8 levels of Classification

1. Domain
2. Kingdom
3. Phylum
4. Class
5. order
6. Family
7. Genus
8. Species

- Bacteria → • prokaryotic cells • unicellular • vital decomposers
- Archaebacteria → • prokaryotic cells • unicellular • live in extreme environments
- Eukaryotes → • Eukaryotic cells • unicellular • Includes Protists, Fungi, Plants, and animals.

Under the Domain Eukarya

- Protists → • unicellular • absorb, photosynthesize, or ingest food.
  • Algae, Protozoans, slime molds, water molds.
- Plants → • multicellular • photosynthesize
  • mosses, Ferns, Conifers, Flowering plants
- Fungi → • multicellular • absorb food
  • molds, mushrooms, yeast, ringworms
- Animals → • multicellular • Ingest Food
  • reptiles, mammals, amphibians, birds, Fishes.

Binomial Nomenclature → "bi" = two, "nomial" = terms, nomenclature = assigning of names.

↳ Each living creature/organism assigned two-part name.

Genus: First word, capitalized, a group of closely related species.

Species: second word, lowercase, a group of similar individuals capable of interbreeding or exchanging genes, producing fertile offspring.

↳ Fertile offspring are able to reproduce.

Speciation: An evolutionary process that results in the formation of a new species.

↳ Modes of speciation:

- Allopatric - due to geographic isolation
- Sympatric - occurs in same location/habitat

Reproductive Isolating Mechanisms:

- Prezygotic - occurs before or prevents fertilization of the egg.
- PostZygotic - occurs after fertilization of the egg.

The Scientific Method

↳ A standard series of steps used to gain new knowledge that is widely accepted among scientists.

- Acts as a guideline for scientific study
- May be modified or adapted to suit a particular field of study.

Steps of the Scientific Method

Observation: Collect and record current information used to construct hypothesis. Includes scientist's own prior knowledge. Can incorporate the knowledge and experience of other scientists.

Hypothesis: possible explanation; An informed statement that can be tested

↳ Three Types:

- Directional: predicts positive or negative change
- Non-Directional: Predicts a change, but is not specific
- Null: predicts no change will occur.

Experimentation: series of procedures designed to collect data for testing a hypothesis. Involves "if, then" logic; searching for cause and effect relationship. Proper design includes testing one specific variable at a time.

Experiment Variables

- Independent Variable: Factor being tested, applied only to experimental group.

· **Dependent Variable**: result of independent variable.
· **Controlled Variables**: all factors not changed

Data/Results: Facts or information collected through experiment. May be presented as tables or graphs. Analyzed using statistics to determine validity.

Conclusion: Data is examined to determine if hypothesis is supported
↳ If prediction occurs, hypothesis is supported.
↳ If prediction does not occur, hypothesis is rejected.

Scientists report their findings in scientific publications, these are peer reviewed to ensure research is credible, accurate, and unbiased.

Scientific Theory vs Scientific Law

Scientific Theory: explains how or why something happens.
↳ · Uses many observations and extensive experimental evidence.
· Can be applied to unrelated facts and new relationships.
· Can be modified if new data/evidence is introduced

Scientific Law: Explains what is happening, often without change.
↳ · Stands the test of time, experimentally confirmed over and over.
· Can create true predictions for different situations
· Uniform and universal